

MORNING

[Total No. of Questions: 09]

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Uni. Roll No. ....

Program: B.Tech. (Batch 2018 onward)

Semester: 3<sup>rd</sup>

Name of Subject: Digital Electronics

Subject Code: ESCS-101

Paper ID: 16012

Scientific calculator is not allowed.

**Time Allowed: 03 Hours**

**Max. Marks: 60**

**NOTE:**

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

**Part – A**

**[Marks: 02 each]**

**Q1.**

- a) Why do we need shift registers?
- b) Identify the fastest logic family along with suitable justification.
- c) Demonstrate the concept of FPGA.
- d) What is Race around condition? How it can be eliminated?
- e) Interpret  $(4096)_{10}$  into its equivalent Gray code and Excess 3 code.
- f) Evaluate  $(11001)_2 \div (101)_2$  using the long division method.

**Part – B**

**[Marks: 04 each]**

- Q2.** Demonstrate the block diagram and working of Programmable Logic Array.
- Q3.** List various types of Digital to Analog converters. Explain R-2R Ladder D/A converter in detail.
- Q4.** Explain the functional block diagram and working of S-R Flip Flop.
- Q5.** Minimize the Boolean expression  $(A+B) (A'+C) (B+C)$  and draw the circuit diagram using NAND gates only.
- Q6.** Design a Johnson or Twisted Ring counter to explain its working states.
- Q7.** Write the expression for Boolean function  $F(A,B,C) = \sum m(1,4,5,6,7)$  in a standard product of sum (POS) form.

## Part – C

[Marks: 12 each]

- Q8. Explain a 3-bit Binary to Gray and 3-bit Gray to binary converter along with their block diagram, truth table and circuit diagram.

OR

Solve the following using K-maps:

a)  $Y = \sum m(3, 5, 6, 11, 12) + \sum d(1, 7, 13, 15)$

b)  $f(A, B, C, D) = \sum m(0, 1, 3, 4, 5, 6, 7, 8, 9) + \Phi(10, 11, 12, 13, 14, 15)$

- Q9. Implement the following function using Type 0, Type 1 and Type 2 of MUX designing.  
 $f(A, B, C, D) = \sum m(0, 1, 3, 4, 6, 8, 12, 14, 15)$ . Comment on changes occur, if would be designed using Type 3.

OR

Explain the following differences:

- (a) A/D converter and D/A converter.
- (b) Moore and Mealy Machines.
- (c) Synchronous and Asynchronous counters.

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